

COLLINS ANYANWU

Geospatial & AI Data Analyst | Database & BI Specialist

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Nationality: Nigerian • Languages: English (Full Professional) • Open to Remote & International Relocation



PROFESSIONAL SUMMARY

Geospatial & AI Data Analyst and Database & BI Specialist with **19+ years** turning spatial and operational data into decisions across the utility, oil & gas, and technology sectors. Manage a **3,800+ km** electricity distribution network serving **1M+ customers**, automated ETL pipelines that cut manual processing by **67%**, and architected dashboards monitoring **20,641+ assets** across 23 data layers. Expert in ArcGIS Enterprise, PostgreSQL/PostGIS, FME, Python (ArcPy, GeoPandas), Power BI, Tableau, and AWS, backed by certified machine-learning and deep-learning training, with full-stack delivery in React, Next.js, TypeScript, and REST APIs. Proven in team leadership, stakeholder management, and Agile delivery of data products.

KEY ACHIEVEMENTS

- Architected an enterprise asset-management platform that gives operations teams a single source of truth for **20,641+** distribution assets across **23** geospatial layers, secured with role-based access control and single sign-on (Next.js, TypeScript, PostGIS).
- Automated GIS workflows with Python (ArcPy, GeoPandas), eliminating **67%** of manual processing effort and cutting mean incident-response time by **40%**.
- Digitized field operations with Survey123 across **8,513+** upriser inspections, accelerating data collection by **45%** and eradicating paper-based errors.
- Directed a team of **8** GIS professionals delivering Tableau and Power BI integrations for utility and infrastructure clients across West Africa.
- Sustained **99.9%** uptime on PostgreSQL/PostGIS production databases and unlocked **50%** query-performance gains through SQL optimization.

CORE COMPETENCIES

Geospatial Data Management • GIS Data Engineering • Spatial Analysis & Modeling • Spatial ETL (FME, ArcPy, GeoPandas) • Enterprise Geodatabase Administration (PostGIS, ArcSDE, Oracle Spatial) • Cloud Infrastructure (AWS - S3, EC2, RDS) • Data Analytics & Business Intelligence (Power BI, Tableau, ArcGIS Dashboards) • Machine Learning & AI • Python Automation • Full-Stack Web Development (React, Next.js, TypeScript, REST APIs) • Data Governance & Quality Assurance • Agile / Scrum Delivery • Stakeholder Management • Team Leadership & Training • Remote Sensing & LiDAR • Field Data Collection (Survey123, GPS/GNSS)

PROFESSIONAL EXPERIENCE

GIS Coordinator

Nov 2017 - Present

Ikeja Electric • Lagos, Nigeria

- Govern the geospatial data backbone of a **3,800+ km** electricity distribution network serving **1M+ customers** across Lagos, engineering Python (ArcPy, GeoPandas) automation that cut manual GIS workflows by **35%**.
- Architected and led the full-stack rebuild of the IE Asset Dashboard (Next.js, TypeScript, PostGIS, REST APIs), giving operations a single source of truth for **20,641+** assets across **23** data layers, secured with role-based access and single sign-on.
- Engineered spatial ETL pipelines (Python, ArcPy, FME) that unify SCADA, billing, and field-capture data, eliminating **67%** of manual data-processing effort.
- Launched real-time ArcGIS outage-monitoring dashboards that cut mean incident-response time by **40%** and now drive executive decision-making for operations leadership.
- Digitized field data capture with Esri Survey123, accelerating collection speed by **45%** and eradicating paper-based errors across field teams.
- Delivered Power BI analytics for transformer-vandalization tracking and **8,513+** upriser inspections, plus a GIS KPIs dashboard (Next.js, Firebase) giving leadership real-time visibility of team and operational performance.

GIS / Data Engineering Manager

Sep 2016 - Oct 2017

PoloSoft Technologies • Lagos, Nigeria

- Directed a team of **8** GIS professionals across concurrent Agile project streams, delivering geospatial solutions on schedule for utility and infrastructure clients across West Africa.
- Championed BI adoption by architecting Tableau and Power BI integrations that enabled client stakeholders to visualize spatial data alongside operational metrics for faster decisions.
- Designed and implemented cloud-based ETL pipelines that scaled large-volume geospatial data processing and warehousing for enterprise clients.
- Built team capability through structured technical training programs, upskilling staff in GIS technologies and database management and raising delivery quality.

GIS / Database Specialist

Oct 2012 - Aug 2016

SpatialMatrix Ltd • Lagos, Nigeria

- Migrated and administered enterprise PostgreSQL/PostGIS databases for oil & gas and infrastructure clients, sustaining **99.9%** uptime across mission-critical production environments.
- Optimized complex spatial SQL queries and indexing strategies, boosting database performance by **50%** for demanding geospatial operations.
- Delivered interactive dashboards, spatial analytics, cartographic production, and database architecture that turned raw spatial data into decision-ready insight for oil & gas, NGO, and infrastructure clients.

Geologist

Jun 2010 - Aug 2012

Earth-Source Hydrocarbon • Nigeria

- Led field geological surveys and GPS-based data collection for hydrocarbon exploration, producing datasets that underpinned resource estimation and prospect appraisal.
- Produced structural and stratigraphic maps that guided drilling-target selection and reserve evaluation for exploration teams.
- Integrated geological field data into ArcGIS for spatial analysis and mapping of prospective zones, translating findings into actionable datasets for exploration and drilling teams.

GIS Technician

Jan 2005 - May 2010

Spatial Technologies Limited • Nigeria

- Built and maintained ETL pipelines for spatial data ingestion, transformation, and quality assurance, ensuring analysis-ready datasets for project teams.
- Performed network, proximity, and spatial-modeling analyses that informed infrastructure planning decisions across telecom and infrastructure projects.
- Managed GPS/GNSS field data-collection campaigns and post-processing for large-scale mapping, while supporting geodatabase maintenance, data migration, and topology validation.

EDUCATION

Masters, Geographic Information Systems

University of Lagos, Lagos, Nigeria

Bachelor of Technology (B.Tech), Geology

Federal University of Technology, Owerri, Nigeria

TECHNICAL SKILLS

GIS & Geospatial: ArcGIS Enterprise, ArcGIS Pro, ArcGIS Online, ArcSDE, QGIS, Survey123, ArcGIS Dashboards, FME (Safe Software), Google Earth Engine, LiDAR, Remote Sensing, GPS/GNSS

Programming: Python (ArcPy, GeoPandas, Shapely, PyQGIS), SQL, R, JavaScript, TypeScript

Data Science & AI: Machine Learning, Deep Learning, Data Analysis, Pandas (certified - Edureka)

Databases: PostgreSQL/PostGIS, Oracle Spatial, SQL Server, Supabase, Firebase

Visualization & BI: Power BI, Tableau, Power Query, Plotly.js, Recharts, Leaflet.js, Matplotlib, Seaborn

Web Development: React, Next.js, Node.js, REST APIs, Tailwind CSS, HTML/CSS, Vercel

Cloud & DevOps: AWS (S3, EC2, RDS), Git/GitHub, Docker, CI/CD (GitHub Actions), Vercel Deployments

CERTIFICATIONS

Python & SQL for Data Science - Edureka, 2024 • Deep Learning Specialization - Edureka, 2024 • Tableau for Data Visualization - Edureka, 2024 • Data Science & Machine Learning Programs - Edureka, 2024 • Big Data Analysis, Power Query, Power Pivot & VBA - UrBizEdge, 2022 • GISP Certification - In Progress (GISCI)